AI NOTES SERIES

AI Governance & Quality Control

Using AI responsibly — testing before trusting, checking work carefully, and keeping people in charge of important decisions.

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These are introductory study notes, written in simple English for teachers, students and beginners. They come from a practical concern I meet every day: checking AI-assisted work before it is trusted and used. Explanations are simplified on purpose, so the main ideas are easy to follow and easy to revise.

What Governance Means

Governance means the rules, habits and checks that keep something under responsible control. In AI, governance is the discipline of using AI carefully and honestly, so that its benefits are real and its mistakes are caught before they cause harm.

Governance is not one tool and not one document. It is a way of working, every time AI is used for something that matters.

> **Plain definition:** AI governance means deciding how AI may be used, testing it before trusting it, watching it while it is used, and keeping a person responsible for the final result.

The parts of good governance

- **Test before trusting.** Try the system on questions with known answers before using its work for real tasks.
- **Measure accuracy.** Do not rely on a feeling that answers look good. Check how often the system is actually right.
- **Watch for bias.** Check whether the system treats different people or groups unfairly.
- **Protect privacy.** Keep personal information out of AI tools unless there is a safe and agreed way to use it.
- **Be transparent about limits.** Say openly what the system can and cannot do, and where its answers may be wrong.
- **Keep human oversight.** A person stays in charge of important decisions, especially decisions that affect people's lives, marks, money or health.

A teacher who checks AI-made answers before marking, a doctor who verifies an AI suggestion before treating a patient, and an office worker who reads an AI draft before sending it are all practising governance, even if they never use the word.

The Central Failure: Hallucination

The most important failure to understand is called **hallucination**. It happens when an AI system produces text that sounds correct and confident, but is not true.

A **language model** is built to produce text that sounds right. It learns patterns in language: which words usually follow which other words. That is how it writes smoothly. But sounding right and being right are two different things.

Why it happens, in simple words

- The model works by continuing patterns. It does not look up facts in the way a person checks a book.
- When its knowledge runs out — on a rare topic, a recent event, or a detail it never learned well — it may still continue the pattern and fill the gap.
- The filled gap can be an invented fact, an invented reference, or an invented figure, written with full confidence and no warning.
- The smooth style hides the problem. A false answer often reads exactly like a true one.

> **Remember:** Confidence is part of the style, not proof of truth. A calm, clear, well-written answer can still be completely wrong.

What to do about it

- **Source-check important claims.** Check names, dates, figures and references against a trusted source before using them.
- **Test on known-answer questions.** Ask questions you already know the answers to, and see how the system performs before trusting it on unknown ones.
- **Prefer systems that can say they do not know.** A system that can admit uncertainty is safer than one that always answers.
- **Treat references with special care.** Book titles, article names and links given by AI should always be verified, because invented references are a common form of hallucination.

Bias

Bias means an unfair pattern: a system leaning towards or against a person or group without a good reason. In AI, bias usually begins in the data.

AI learns patterns from data. If the data holds unfair patterns — because the past was unfair, or because some groups appear in the data much more than others — the system can learn those patterns and repeat them in its outputs. The system does not intend to be unfair. It simply repeats what it found.

A simple example

Imagine a tool that helps a company read job applications. If, in the past, the company mostly hired people from one group, the old records will show that pattern. A tool trained on those records may learn to favour applications from the same group and push others down — even if nobody wrote such a rule, and even if better candidates are being missed. The unfairness of the past has quietly become the behaviour of the tool.

Why testing with different groups matters

- A system can look accurate overall while still failing one group badly. Testing only as a whole can hide that failure.
- Testing with different groups — for example different ages, genders, languages or regions — shows whether results stay fair for everyone.
- Language is a clear case: a tool may work well in a widely used language and poorly in a less common one, because it saw far less data in that language during training.
- When unfair results are found, the honest response is to fix the data, adjust the system, or limit its use — not to hide the result.

Privacy

Privacy means keeping personal information about real people safe and under control. When we type or paste something into an AI tool, that material leaves our own notebook or computer and goes to a service run by others. We should act as if anything we paste could be stored, and choose carefully what we share.

What should not be pasted into AI tools

- Full names together with identity card numbers, passport numbers or bank details.
- Passwords, PIN codes, or secret codes of any kind.
- Private phone numbers, home addresses and personal email addresses of other people.
- Medical details that identify a patient, or private family matters about real people.
- Exam papers, answer keys or confidential office documents before they are officially released.

Care with other people's data

- **Students' data.** A teacher should not paste a class list with names, marks or behaviour notes into a general AI tool. Remove names first, or work with made-up examples instead.
- **Patients' data.** Health workers carry a special duty. Identifiable patient information belongs in approved health systems, not in general chat tools.
- **Customers' data.** Names, orders and complaints from customers were shared for a service, not for pasting into outside tools.

> **Safe habit:** Before pasting, ask one question — would the person this information belongs to agree to it being shared here? If the answer is no or unsure, do not paste it. Change names and remove details, or do the task without the AI tool.

Quality Control in Practice

Quality control means checking work against a standard before it is accepted and used. With AI, quality control turns governance from an idea into daily practice. These are the habits that matter most.

Before the system is used

- **Evaluate on a known-answer test set.** Prepare a set of questions or tasks where the correct answers are already known. Run the system on that set and count how many it gets right, and where it fails.
- **Set clear rules.** Write down what the AI may do, what it may not do, and which tasks always need a human check. Clear rules prevent slow drift into unsafe use.

While the system is used

- **Check a sample of real outputs by hand.** Nobody can check every output, but regularly reading a sample closely shows the real quality, including mistakes the test set missed.
- **Keep records of what was checked.** A short record — what was tested, when, what was found, and what was changed — makes the work honest and repeatable, and helps the next person who takes over.
- **Keep a human as the final decision-maker.** For anything that affects people — marks, selection, money, health, discipline — the AI may suggest, but a person decides and a person is responsible.

> **Simple rule:** The more a decision affects a person's life, the stronger the human check must be. Low-risk drafts can be checked lightly. High-risk decisions should never rest on AI output alone.

Honest Limits of Governance

Good governance matters, but it has honest limits. Saying those limits clearly is part of governance itself, because false confidence is one of the risks we are trying to control.

- **Evaluation can only test what someone thought to test.** A test set covers the questions its maker imagined. Real life will always produce questions, accents, situations and edge cases that nobody imagined in advance.
- **Governance reduces risk; it does not certify perfection.** A system that passed every check can still make a serious mistake tomorrow. A passed test means risk was reduced, never that risk is zero.
- **Checking must continue after launch.** Data changes, users change, and the system itself may be updated by its maker. A check done once, long ago, says little about the system today.
- **Records only help if someone reads them.** Keeping records is not the goal. Reviewing them, learning from the mistakes they show, and changing practice is the goal.

> **Honest summary:** Governance is not a stamp that says a system is safe forever. It is a continuing habit — test, check, record, review, and stay ready to stop or fix the system when the evidence says so.

This is why these notes return again and again to the same simple practice: check the work. In my own daily work, AI helps with drafting, organising and preparing material, but the checking — reading outputs carefully, verifying facts, and deciding what is good enough to use — stays human work. That division is not a weakness of the method. It is the method.

Words to Know

Governance — the rules, habits and responsibilities that keep the use of AI under careful human control.

Quality control — checking outputs against a standard before they are accepted and used, so mistakes are caught early.

Hallucination — when an AI system produces false information written in a confident, believable style, such as an invented fact, reference or figure.

Bias — an unfair pattern in a system's behaviour, often learned from unfair or unbalanced patterns in its training data.

Evaluation — the organised testing of a system to measure how well it performs and where it fails.

Test set — a prepared set of questions or tasks with known correct answers, used to check a system fairly before trusting it.

Oversight — human watching and control over a system, including the power to correct it, limit it or stop it.

Transparency — being open about what a system does, what data it uses, and what its limits are, so people can judge it honestly.

Privacy — keeping personal information about real people safe, shared only with care and with a proper reason.

Quick Revision

- Governance is the discipline of using AI responsibly: test before trusting, measure accuracy, watch for bias, protect privacy, be open about limits, and keep human oversight.
- Hallucination is the central failure: language models produce text that sounds right, and may invent facts, references or figures with full confidence.
- Confidence is style, not proof. Source-check important claims and verify every reference.
- Bias begins in data: unfair patterns in data can become unfair outputs, so test with different groups.
- Never paste passwords, identity numbers, or identifiable information about students, patients or customers into general AI tools.
- Quality control in practice: evaluate on known-answer test sets, check a sample of real outputs by hand, keep records, and set clear rules for what AI may and may not do.
- A human stays the final decision-maker for anything that affects people.
- Governance reduces risk but does not certify perfection, and checking must continue after launch, not stop.

Closing Note

These notes are an introduction to governing AI and controlling its quality. Explanations are simplified on purpose, so a beginner can build clear habits first and add detail later. They are study notes from my own learning and from the practical work of checking AI-assisted material every day, shared for students, teachers and anyone who wants to use AI carefully and honestly.

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